Fiscal Year:	FY 2025	**Task Last Updated:**	FY 09/16/2024
PI Name:	Crucian, Brian Ph.D.		
Project Title:	Pilot Assessment of Stress and Latent Herpesvirus Reactivation at Palmer, Antarctica – Platform for Validation of Immune Countermeasures?		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Immune**:Risk of In Mission Impacts, Adverse Health Events or Long-Term Health Impacts due to Altered Immune Response		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	brian.crucian-1@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	281-483-7061
Organization Name:	NASA Johnson Space Center		
PI Address 1:	Immunology, SK4		
PI Address 2:	2101 NASA Pkwy		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058-3607	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	12/01/2018	**End Date:**	10/31/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/31/2022 per PI (Ed., 4/23/21)		
Key Personnel Changes/Previous PI:	Per the PI, Douglass Diak and Cody Gutierrez were added as Co-Investigators (Ed., 10/14/24).		
COI Name (Institution):	Mehta, Satish Ph.D. (JES Tech/NASA Johnson Space Center) Makedonas, George Ph.D. (JES Tech/NASA Johnson Space Center) Krieger, Stephanie B.S. (KBRwyle/NASA Johnson Space Center) Bhattacharya, Sharmila Ph.D. (NASA Ames Research Center) Paul, Amber Ph.D. (NASA Ames Research Center) Diak, Douglass (Aegis Aerospace/ NASA Johnson Space Center) Gutierrez, Cody (JES Tech/NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Recent publications have characterized adverse health events potentially related to immune system dysregulation in some crewmembers onboard the International Space Station (ISS). An appropriate ground analog has yet to be validated, although several European bases are being characterized. This study seeks to collect low-cost pilot data assessing stress, immunity, and viral reactivation during winterover at the U.S. Palmer Station, Antarctica. The goal is to ascertain if Palmer may serve as a spaceflight analog option for ground validation of immune countermeasures. NASA currently has no relevant data from Palmer station. This study may be performed by simply returning frozen saliva samples, frozen plasma from a venous blood sample, a 3 cm hair sample, as well as a rapid fingerstick blood analysis on location from crewmembers performing winterover. The inclusion of a hand held fingerstick blood analyzer will enable the use of a neutrophil/lymphocyte ratio as an indicator of disease susceptibility. Returned saliva samples will be used to assess a variety of parameters including stress hormones, cytokines/inflammation, and latent herpesviruses (an excellent flight-validated biomarker of immune dysregulation). Returned plasma samples will be used to assess plasma cytokines/inflammation biomarkers. Aims 1. Investigate the effect of coastal Antarctica winterover on salivary stress hormones, salivary cytokine profiles, plasma cytokines, and basic peripheral leukocyte distribution. 2. Investigate the effect of coastal Antarctica winterover on the reactivation and shedding of latent herpesviruses. 3. Characterize other adverse clinical outcomes, as voluntarily shared by crewmembers via a health survey, such that immune changes, viral reactivation, and clinical manifestations may be correlated to infer information regarding clinical risk from persistent immune dysregulation.
Rationale for HRP Directed Research:	The justification for Directed Task is the highly constrained nature of this pilot study involving focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal, which could include a follow on task to a solicitation. This is a low-cost pilot study that would have been classified as Discretionary if not for the use of an analog site (Antarctica).
Research Impact/Earth Benefits:	This study may be able to provide new insights into the relationship between immune function, stress, and adverse clinical events. In particular, as relates to confinement and deployment stress.
Task Progress:	The 2021 winter-over season concluded with samples being shipped from Palmer Station, Antarctica to NASA Johnson Space Center early 2022. All samples – including saliva, plasma, health surveys, etc. – have since been processed as of early 2023. Data and statistical analysis were conducted and the findings of the study were published in the Life Sciences in Space Research Journal in August of 2023. In summary, the research team was able to conclude, given the data, that winter-over at Palmer Station, Antarctica was analogous to spaceflight in regards to the alterations seen in the immune system of both astronauts and Antarctica volunteers. These findings will permit further research to be done in Palmer Station to test potential immune countermeasures for spaceflight.
Bibliography Type:	Description: (Last Updated: 05/21/2026)
Articles in Peer-reviewed Journals	Diak DM, Krieger S, Gutierrez C, Mehta S, Nelman-Gonzalez M, Babiak-Vazquez A, Young M, Oswald TM, Choukér A, Johnson J, James H, Chang CY, Crucian B. "Palmer Station, Antarctica: A ground-based spaceflight analog suitable for validation of biomedical countermeasures for deep space missions" Life Sci Space Res (Amst). 2024 Feb;40:151-57. Epub 2023 Aug 17. https://doi.org/10.1016/j.lssr.2023.08.001 ; PMID: 38245340 , Feb-2024